

Munich, Germany
German citizen (EU)
+49 176 42997580
yannick.burkhardt@web.de
yannickburkhardt.github.io

Skills

- C++, Python
- PyTorch, ONNX
- Eigen, OpenCV, Open3D
- GPU inference optimization: TensorRT, quantization, CUDA graphs
- ROS, ROS 2, DDS
- Docker, Linux, Git

Languages

- German: native
- English: fluent
- Spanish
- Polish

Interests

- Event-based vision, visual-inertial SLAM
- Embodied AI, learned perception
- Path planning, robot control, manipulation
- Hiking, football, water polo, chess
- Travel, intercultural experiences

Yannick Burkhardt

Robotics researcher who takes learned perception from paper to real-time systems on real robots. PhD on machine learning for event-based visual-inertial SLAM at TUM & ETH Zürich; previously co-developed a manipulator control stack at Agile Robots.

Education

Apr. 2024 – Present (expected end of 2027) — **Technical University of Munich & ETH Zürich (Mobile Robotics Lab)**
PhD on event-based robot perception
Advisor: Prof. Stefan Leutenegger
- *SuperEvent*: learned event-camera keypoints; 45% lower trajectory error than the prior best
- *AERO-VIS*: first purely event-based VI-SLAM to fly a UAV in closed loop, fully onboard
- Supervised 3 master's theses (4th ongoing); reviewer for T-RO, RA-L, ICRA, IROS

Feb. – May 2027 — Research visit, ACFR, Univ. of Sydney (upcoming)

Jul. 2025 — Int. Computer Vision Summer School (ICVSS)

Oct. 2020 – Jul. 2023 — **Technical University of Munich**
M.Sc. Robotics, Cognition, Intelligence
Grade: 1.1 (German scale, 1.0 = best)

Feb. – Jun. 2023 — Tecnológico de Costa Rica (exchange semester)

Aug. 2022 — Int. Elite Robotics Summer School (SDU, Denmark)

Oct. 2016 – Feb. 2020 — **Karlsruhe Institute of Technology**
B.Sc. Mechatronics and Information Technology
(Grade: 1.6)

Selected Publications

Y. Burkhardt, S. B. Laina, S. Boche, L. Freißmuth, S. Leutenegger. *AERO-VIS: Asynchronous Event-based Real-time Onboard Visual-Inertial SLAM*. RA-L 2026.

Y. Burkhardt, S. Schaefer, S. Leutenegger. *SuperEvent: Cross-Modal Learning of Event-based Keypoint Detection for SLAM*. ICCV 2025 (**Highlight**, top 2.5% of submissions).

Y. Burkhardt*, Q. Feng*, J. Feng, K. Sharma, Z. Chen, A. Knoll. *Multi-fingered Dynamic Grasping for Unknown Objects*. Humanoids 2024.

B. Varga, **Y. Burkhardt**, S. Schwab, S. Hohmann. *Shared-Control Concepts for Large Vehicle-Manipulators*. ISIE 2020.

*equal contribution

Awards & Talks

- Winner (event-only track): IROS 2025 Event-based SLAM Challenge (*SuperEvent*)
- Keynote talk: ICCV 2025 Workshop on Multi-modal Localization & Mapping
- PhD scholarship: Munich Center for Machine Learning (MCML), 2024
- M.Sc. with high distinction (2023): top 1.5% of graduates
- Deutschlandstipendium (national merit scholarship), 5×, 2017–2023

Work Experience

Aug. 2023 – Feb. 2024 — **Agile Robots, Munich**: Robot Software Engineer
Multi-body control for a 6-DoF manipulator; co-developed a new control stack from scratch

Oct. 2020 – Jan. 2023 — **Agile Robots**: Working Student & Master's Thesis
Visual servoing for learned grasping with a five-fingered hand (*Humanoids 2024*); kinematic calibration

Oct. 2019 – May 2020 — **LEONI Elocab Ltd., Canada**: Internship
Cable construction and testing

Oct. 2018 – Aug. 2019 — **FZI Research Center for Information Technology**: Research Assistant & Bachelor's Thesis
Shared control for vehicle-manipulators (*ISIE 2020*)